Murray State University

Types of AI

Demo 1 - model representations and learning methods - 20 minutes

You need: one browser and the demo's `index.html`. It works offline. Start with a fresh page; use Settings -> Presentation mode to project, Open Presenter Notes for this guide in the app, and Open printable PDF when a handout helps.

The question for all 20 minutes

How is a strategy represented, and how did it get there? A-C compare written rules, a value table, and shared neural-network weights on one game. D introduces other model families. The separate **How they learn** tab explains training methods.

> *"A model is how the strategy is represented. A learning method is how its adjustable parts change."*

1a - Symbolic AI (GOFAI) - 4 minutes

Question

Can an opponent be skilled when it has never learned from a game?

Activity

1. Open **1a Symbolic AI (GOFAI)** and play a few moves against All 8.
2. Point to the highlighted rule after each reply. Switch to First 2 and invite the room to find a fork, then return to All 8.
3. Open **Details** for the exhaustive policy evidence and the explanation of GOFAI.

Expected observation

The opponent can explain a move by naming a rule. All 8 has no losing line because the app checks the whole policy, not because it won a few demonstrations.

Teaching limit

GOFAI means Good Old-Fashioned Artificial Intelligence. These rules form a decision list: a matching rule chooses a move; otherwise the next rule is checked. It can be drawn as a decision tree. Here people wrote the tests; trees can also be learned.

1b - Value table - 5 minutes

Question

What changes when an opponent keeps a score for positions it has experienced?

Activity

1. Open **1b Value table**. In Era 0, use **Show move scores**; every score begins at zero.
2. Play one short game, then run a visible training burst. Select an earlier and a later era to compare their move scores.
3. Use **Details** to inspect evidence for the selected era, then return to play.

Expected observation

The model is a table; the learning method is reinforcement learning. Game outcomes and estimates of later replies update separate position scores. Practice can therefore change its choices.

Teaching limit

Human play does not train this table. Only the app's training burst updates it. The table also has one separate score per position, so it does not automatically carry what it learned to a different situation.

1c - Neural Network - 7 minutes

Question

Can adjustable numbers reused across many boards approximate the scores a table learned?

Activity

1. Open **1c Neural Network** and choose **Create learning examples**. A fresh table learner plays 20,000 seeded games; its estimates become frozen teaching targets.
2. Inspect the network's layers and a connection's numeric weight. Choose **Train network**, then compare the weight with its initial value.
3. Compare training error, held-out error, and the separate playing score. Use **Show move scores** to inspect predictions for legal moves.

Expected observation

This feedforward multilayer perceptron uses supervised learning. Backpropagation calculates how weights affect error; normalized gradient updates adjust them. Shared weights influence many positions. Neuron activations are responses to an input, not stored weights; a positive weight does not mean a good move.

Teaching limit

Lower error means closer agreement with the teacher, not guaranteed playing strength. People chose the inputs and learning procedure. During play the network calculates its own scores without consulting the frozen table. It is trained with gradients, not evolutionary selection.

A fresh teacher can leave an unvisited position at its initial score of zero. That is a limit of the examples, not proof that the position is neutral in perfect tic-tac-toe.

> *"A neural network is a type of machine learning. Here it learns a compact approximation of another learner's experience."*

Other models and how they learn - 4 minutes

D - Other model types introduces linear models, decision trees, ensembles, nearest neighbors, support-vector machines, and probabilistic models. Search/planning is explained separately as a problem-solving approach.

Open the unlettered **How they learn** tab. Supervised learning uses example targets; unsupervised learning finds structure without target labels; self-supervised learning creates targets from the data; reinforcement learning uses action rewards; semi-supervised learning combines labeled and unlabeled examples. These approaches can be combined.

Self-play is not the same as self-supervision. B uses reinforcement learning; C uses supervised targets produced by B's algorithm. A neural network can use other learning methods too.

Close with evidence, not a winner

- Which model stores separate values, and which shares weights?
- Why does lower held-out error not prove unbeatable play?

Optional Ultimate tic-tac-toe explores missing inputs: shared weights cannot supply omitted facts about a board's place in the larger game or where a move sends the opponent.

Reopen Guide with the header's Guide button. Settings also offers Open Presenter Notes, Presentation mode, and Reset; Reset returns the full demonstration to its initial state.

Bryant Harrison - Murray State University - Types of AI - single-file offline demo - no accounts or runtime network requests